HealthPulse AI

BMI Detection and Metabolism Analysis System Using Full Stack Web Technologies

Project Report

Submitted by

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Certificate

This is to certify that the project report entitled **"HealthPulse AI: BMI Detection and Metabolism Analysis System Using Full Stack Web Technologies"** submitted by **Nandana MC** in partial fulfillment of the requirements for the award of the degree in Computer Science and Engineering is a bonafide record of the work carried out under our supervision.

Project Guide | Head of Department

Declaration

I hereby declare that this project report entitled **"HealthPulse AI: BMI Detection and Metabolism Analysis System Using Full Stack Web Technologies"** is a record of original work carried out by me under the guidance of the project supervisor.

Nandana MC

Acknowledgement

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Abstract

The rapid increase in obesity, unhealthy lifestyles, metabolic disorders, and chronic diseases has created a significant demand for intelligent healthcare monitoring systems capable of providing accurate health analysis and preventive healthcare support. Traditional healthcare applications and BMI calculators often provide only basic obesity classification without offering comprehensive metabolism analysis, calorie estimation, personalized recommendations, or real-time healthcare analytics. These limitations reduce their effectiveness in helping users maintain healthy lifestyles and prevent obesity-related medical complications.

This project presents "HealthPulse AI," an intelligent Full Stack web-based healthcare platform developed for automated BMI detection, metabolism analysis, calorie estimation, and AI-driven personalized healthcare recommendation generation. The proposed system integrates modern Full Stack Web Technologies including React.js, Node.js, Express.js, and MongoDB with Artificial Intelligence-based healthcare analytics to develop a responsive, scalable, secure, and user-friendly healthcare application. The platform enables users to enter healthcare parameters such as height, weight, age, gender, and physical activity level for performing comprehensive health analysis.

The system calculates Body Mass Index (BMI), Basal Metabolic Rate (BMR), daily calorie requirements, and obesity risk levels using standard healthcare formulas and intelligent analytical methods. Based on user health conditions and metabolism analysis, the AI recommendation engine generates personalized diet plans, exercise routines, hydration reminders, calorie management suggestions, and healthy lifestyle guidance. Interactive dashboards and graphical healthcare analytics improve user understanding and support continuous health monitoring and fitness management.

The frontend of the proposed system is developed using React.js, HTML, CSS, JavaScript, and Bootstrap to provide responsive user interfaces suitable for desktop and mobile devices. The backend services are implemented using Node.js and Express.js for API management, healthcare calculations, and recommendation processing. MongoDB cloud database services are used for secure healthcare data storage and real-time data accessibility. Security mechanisms including JWT authentication, encrypted passwords, and protected API routes ensure healthcare data confidentiality and secure user management.

Experimental analysis demonstrates that the proposed HealthPulse AI system provides accurate BMI calculation, efficient metabolism analysis, fast healthcare data processing, and effective personalized recommendation generation. The system improves preventive healthcare awareness, enhances user engagement, and supports intelligent fitness monitoring through AI-driven healthcare analytics and cloud-based healthcare management.

The proposed platform can be effectively implemented in hospitals, healthcare organizations, fitness centers, wellness applications, educational institutions, telemedicine systems, and personal healthcare management environments. Future enhancements such

as wearable device integration, IoT-based healthcare monitoring, AI chatbot assistance, disease prediction systems, and mobile healthcare applications can further improve the intelligence, scalability, and functionality of the platform.

Therefore, the proposed “HealthPulse AI” system provides an efficient, scalable, secure, and intelligent solution for modern digital healthcare applications and preventive healthcare management.

Contents

Chapter 1

Introduction

In recent years, obesity, unhealthy lifestyles, and metabolic disorders have become major healthcare concerns worldwide. Millions of individuals suffer from chronic diseases due to improper health monitoring and lack of preventive healthcare awareness.

Body Mass Index (BMI) is one of the most commonly used indicators for identifying obesity levels. Metabolism analysis helps estimate energy consumption and calorie requirements. Traditional healthcare systems often lack intelligent recommendation capabilities and real-time healthcare analytics.

Artificial Intelligence and Full Stack Web Technologies provide opportunities for developing intelligent healthcare platforms capable of automating healthcare analysis and improving user interaction.

The proposed HealthPulse AI system integrates Artificial Intelligence and Full Stack Web Technologies to provide automated BMI detection, metabolism analysis, calorie estimation, and personalized healthcare recommendations.

Body Mass Index (BMI) is one of the most widely used healthcare indicators for identifying body fat levels and obesity categories based on a person's height and weight. BMI classification helps determine whether an individual is underweight, normal weight, overweight, or obese. Similarly, metabolism analysis is used to estimate the body's energy consumption rate and calorie requirements for performing daily physiological activities such as breathing, blood circulation, digestion, and physical movement. Basal Metabolic Rate (BMR) calculations help estimate the minimum energy required by the body to maintain normal biological functions.

Traditional healthcare systems and BMI calculators generally involve manual calculations, limited personalization, and lack of intelligent healthcare analytics. Most existing healthcare applications provide only simple BMI calculations without offering comprehensive metabolism analysis, calorie estimation, AI-based recommendations, or real-time healthcare monitoring. In many cases, users must use separate applications for calorie tracking, fitness planning, hydration monitoring, and diet management, resulting in poor user experience and inefficient healthcare management. Additionally, several existing systems lack secure healthcare data management, cloud-based accessibility, responsive user interfaces, and personalized healthcare support.

The proposed "HealthPulse AI" system is an intelligent Full Stack healthcare platform developed for automated BMI detection, metabolism analysis, calorie estimation, and AI-driven personalized healthcare recommendation generation. The system allows users to enter healthcare-related parameters such as height, weight, age, gender, and physical activity level. Based on these inputs, the platform calculates Body Mass Index (BMI), Basal Metabolic Rate (BMR), daily calorie requirements, and obesity risk levels using healthcare formulas and analytical techniques.

The system also integrates Artificial Intelligence-based recommendation modules capable of generating personalized diet plans, exercise routines, hydration reminders, calorie management suggestions, and healthy lifestyle recommendations. Interactive healthcare dashboards and graphical analytics improve user understanding and encourage regular healthcare monitoring. The proposed platform aims to increase healthcare awareness, improve fitness management, and support preventive healthcare through intelligent healthcare analytics and real-time monitoring.

The frontend of the proposed HealthPulse AI system is developed using React.js, HTML, CSS, JavaScript, and Bootstrap to create responsive and user-friendly interfaces. Backend services are implemented using Node.js and Express.js for healthcare calculations, recommendation processing, API management, and secure communication. MongoDB Atlas cloud database services are used for storing user healthcare records, BMI history, metabolism reports, and personalized recommendations securely.

The proposed system also implements multiple security mechanisms including JWT authentication, password encryption, protected API routes, secure cloud database access, and input validation to ensure healthcare data confidentiality, integrity, and secure user management. Cloud-based deployment technologies further improve scalability, availability, and remote healthcare accessibility.

The HealthPulse AI platform provides several advantages including automated healthcare analysis, AI-based recommendation generation, real-time health monitoring, secure cloud-based healthcare management, responsive web interfaces, and interactive healthcare analytics dashboards. The system can be effectively implemented in hospitals, healthcare organizations, fitness centers, educational institutions, wellness applications, telemedicine systems, and personal healthcare management environments.

Furthermore, the proposed platform can be extended in the future with advanced technologies such as IoT-based healthcare monitoring, wearable device integration, Artificial Intelligence chatbots, disease prediction systems, Blockchain-based healthcare security, mobile healthcare applications, and Deep Learning-based healthcare analytics. These future enhancements can transform the proposed system into a comprehensive intelligent healthcare ecosystem capable of supporting modern digital healthcare transformation and preventive healthcare management.

Therefore, the proposed "HealthPulse AI: BMI Detection and Metabolism Analysis System Using Full Stack Web Technologies" provides an intelligent, scalable, secure, and efficient solution for automated healthcare analysis, personalized healthcare recommendations, and preventive healthcare monitoring in modern digital healthcare environments.

Chapter 2

Problem Statement

Traditional BMI calculators and healthcare monitoring systems often provide only limited functionalities such as basic BMI calculation without offering comprehensive metabolism analysis, personalized recommendations, calorie tracking, or intelligent healthcare analytics.

Most existing healthcare applications fail to provide:

- Real-time health monitoring
- AI-based recommendations
- Personalized fitness analysis
- Secure cloud-based healthcare management
- Interactive healthcare dashboards

Therefore, there is a need for an intelligent Full Stack healthcare platform capable of automated BMI detection and AI-driven healthcare recommendation generation.

Chapter 3

Objectives

The main objectives of the proposed system are:

- To develop an automated BMI detection system
- To perform metabolism analysis and BMR calculation
- To estimate daily calorie requirements
- To generate AI-based personalized recommendations
- To provide real-time healthcare monitoring
- To develop a responsive Full Stack healthcare application
- To ensure secure healthcare data management

Chapter 4

Literature Survey

Several researchers have developed healthcare systems for BMI prediction, metabolism analysis, and personalized healthcare support.

AI-based healthcare systems analyze user parameters such as body weight, height, calorie intake, and activity levels to provide predictive healthcare insights.

Existing BMI detection systems mainly provide obesity classification based on standard BMI formulas. However, many current systems lack:

- Comprehensive metabolism analysis
- AI-based recommendation generation
- Secure cloud-based healthcare management
- Real-time healthcare analytics

The proposed HealthPulse AI system integrates BMI detection, metabolism analysis, AI recommendation generation, and healthcare dashboards into a single intelligent platform.

The rapid growth of digital healthcare technologies, Artificial Intelligence, cloud computing, and Full Stack Web Development has significantly transformed healthcare monitoring and preventive healthcare systems. Researchers across the world have focused on developing intelligent healthcare applications capable of monitoring obesity levels, metabolism rates, calorie consumption, physical fitness, and lifestyle behaviors. Several healthcare systems have been proposed for Body Mass Index (BMI) detection, metabolism analysis, personalized healthcare recommendation generation, and real-time healthcare monitoring. This chapter presents a detailed review of existing research works and technologies related to healthcare monitoring systems, BMI analysis, metabolism tracking, and AI-based healthcare platforms.

4.1 AI-Based Healthcare Monitoring Systems

Artificial Intelligence has become one of the most important technologies in modern healthcare applications. Researchers have proposed several AI-powered healthcare systems capable of analyzing user health conditions, predicting obesity risks, and generating personalized healthcare recommendations. These systems use Machine Learning algorithms, predictive analytics, and healthcare datasets to automate health analysis and improve healthcare decision-making.

AI-based healthcare systems generally collect user parameters such as:

- Height
- Weight
- Age
- Gender
- Physical Activity Levels
- Calorie Intake
- Lifestyle Habits

Based on these parameters, AI algorithms generate healthcare insights and fitness recommendations. Some advanced healthcare systems also integrate predictive analytics for disease risk detection and obesity prediction.

Advantages

- Automated healthcare analysis
- Personalized recommendations
- Improved healthcare accessibility
- Real-time health monitoring
- Better preventive healthcare support

Limitations

- Limited healthcare dataset availability
- Reduced prediction accuracy for diverse populations
- High computational complexity
- Security and privacy concerns

4.2 BMI Detection and Obesity Analysis Systems

Body Mass Index (BMI) is one of the most commonly used healthcare indicators for obesity classification and fitness analysis. Several web-based and mobile healthcare applications have been developed for automated BMI calculation and obesity monitoring.

Traditional BMI systems classify users into categories such as:

- Underweight
- Normal Weight
- Overweight

- Obese

These systems generally calculate BMI using standard healthcare formulas based on height and weight measurements. Some modern applications also provide graphical health reports and basic calorie tracking functionalities.

Several healthcare researchers have proposed intelligent obesity analysis systems capable of identifying obesity-related risks and providing fitness management guidance. However, many existing systems focus only on BMI classification without performing comprehensive metabolism analysis or AI-based healthcare recommendation generation.

Advantages

- Easy obesity identification
- Simple healthcare analysis
- Fast BMI calculation
- Increased health awareness

Limitations

- No metabolism analysis
- Lack of personalized healthcare support
- No intelligent recommendation systems
- Limited healthcare visualization

4.3 Metabolism Analysis and Calorie Tracking Systems

Metabolism analysis systems estimate the energy consumption rate of the body and calculate daily calorie requirements. Several healthcare applications and fitness platforms provide Basal Metabolic Rate (BMR) analysis and calorie estimation features for weight management and nutritional planning.

These systems generally analyze:

- Age
- Gender
- Height
- Weight
- Physical Activity Levels

Based on these parameters, the systems estimate daily calorie requirements for:

- Weight Loss

- Weight Maintenance
- Muscle Gain
- Balanced Nutrition

Some advanced systems integrate fitness tracking and dietary monitoring features. However, several current healthcare platforms lack AI-based healthcare analytics, secure cloud-based storage, and integrated healthcare dashboards.

Advantages

- Personalized calorie estimation
- Better dietary planning
- Improved fitness management
- Metabolism monitoring support

Limitations

- Limited healthcare personalization
- Lack of integrated healthcare systems
- No AI-driven recommendation generation
- Reduced scalability

4.4 Full Stack Web-Based Healthcare Applications

The development of Full Stack Web Technologies has enabled researchers to build responsive and scalable healthcare applications capable of supporting real-time healthcare monitoring and cloud-based healthcare management.

Modern healthcare platforms commonly use technologies such as:

- React.js
- Node.js
- Express.js
- MongoDB
- Bootstrap
- REST APIs

React.js is widely used for frontend development because of its component-based architecture and dynamic rendering capabilities. Node.js and Express.js are commonly used for backend API development and real-time data processing. MongoDB provides cloud-based NoSQL database management suitable for storing healthcare records and analytics data.

Several researchers have proposed Full Stack healthcare applications for online healthcare monitoring, telemedicine services, digital healthcare management, and patient analytics. These applications improve healthcare accessibility and support cloud-based healthcare services.

Advantages

- Responsive user interfaces
- Cloud-based accessibility
- Real-time healthcare management
- Scalable healthcare architecture
- Interactive dashboards

Limitations

- Security vulnerabilities
- Complex backend management
- High server-side processing requirements
- Limited AI integration

4.5 AI-Based Recommendation Systems

Recommendation systems are widely used in healthcare applications for generating personalized healthcare suggestions and fitness guidance. AI-based recommendation engines analyze user behavior, health conditions, fitness goals, and metabolism analysis results to provide customized healthcare recommendations.

These systems can generate:

- Diet Plans
- Exercise Recommendations
- Hydration Reminders
- Sleep Improvement Suggestions
- Healthy Lifestyle Guidance

Machine Learning and Artificial Intelligence algorithms improve healthcare personalization and user engagement. Some systems also integrate predictive healthcare analytics for identifying obesity-related risks and chronic disease possibilities.

Advantages

- Personalized healthcare guidance
- Improved user engagement
- Better fitness management
- Automated recommendation generation

Limitations

- Dependency on healthcare datasets
- Reduced recommendation accuracy
- Privacy concerns
- Lack of adaptive healthcare learning

4.6 Cloud-Based Healthcare Systems

Cloud computing technologies have significantly improved healthcare data accessibility, scalability, and storage management. Cloud-based healthcare systems allow users to store healthcare records securely and access healthcare services remotely using internet-connected devices.

Cloud healthcare platforms support:

- Remote healthcare monitoring
- Secure healthcare data storage
- Multi-device accessibility
- Real-time healthcare synchronization
- Scalable healthcare services

Several modern healthcare systems integrate cloud databases such as MongoDB Atlas and cloud deployment services for improving healthcare availability and performance.

Advantages

- Improved healthcare accessibility
- Secure cloud storage
- Remote healthcare services
- Better scalability
- High availability

Limitations

- Data security challenges
- Internet dependency
- Privacy concerns
- Cloud maintenance complexity

4.7 Research Gap Identification

Although several healthcare monitoring systems and BMI analysis applications have been developed, many existing systems focus only on individual functionalities such as BMI calculation or calorie tracking. Most current applications fail to integrate BMI detection, metabolism analysis, AI-based recommendation generation, secure cloud-based healthcare management, and interactive healthcare dashboards within a single intelligent healthcare platform.

Additionally, several existing healthcare systems lack:

- Personalized healthcare analytics
- Intelligent recommendation generation
- Secure Full Stack healthcare implementation
- Real-time healthcare monitoring
- Scalable cloud-based healthcare management
- Interactive healthcare visualization

To overcome these limitations, the proposed "HealthPulse AI" system integrates Artificial Intelligence, Full Stack Web Development, cloud database management, BMI detection, metabolism analysis, calorie estimation, and healthcare recommendation generation into a unified intelligent healthcare platform. The proposed system aims to improve preventive healthcare awareness, healthcare accessibility, user engagement, and personalized healthcare monitoring using modern digital healthcare technologies.

Chapter 5

Proposed System

The proposed HealthPulse AI system is an intelligent healthcare platform designed for:

- Automated BMI Detection
- Metabolism Analysis
- Calorie Estimation
- Personalized Healthcare Recommendations

Users enter:

- Height
- Weight
- Age
- Gender
- Activity Level

The system calculates:

- BMI
- BMR
- Daily Calorie Requirements
- Obesity Risk Levels

The AI recommendation engine generates personalized fitness and diet recommendations.

Chapter 6

System Architecture

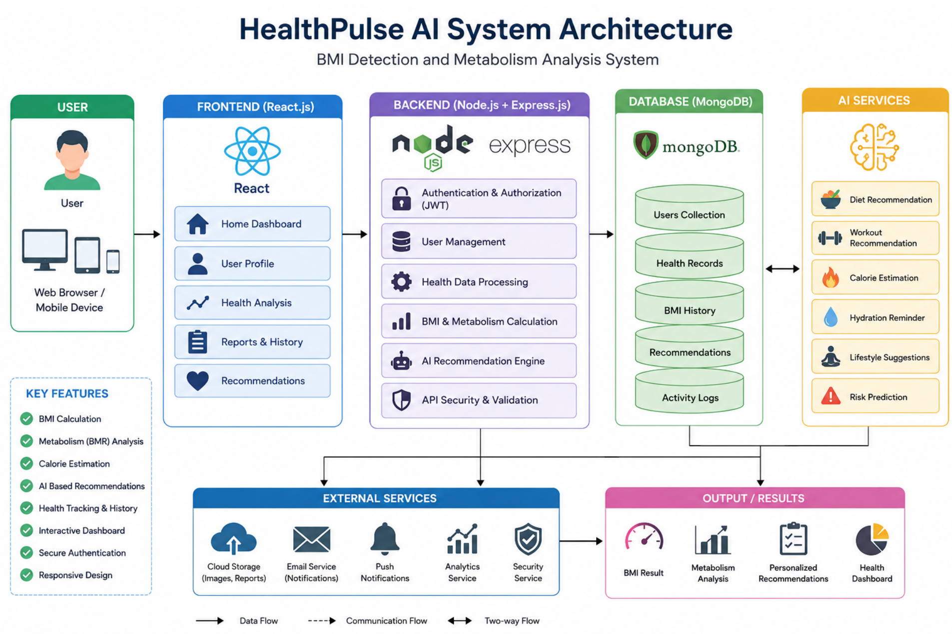

Figure 6.1: HealthPulse AI System Architecture

The architecture contains:

- User Interface
- Authentication Module
- BMI Detection Engine
- Metabolism Analysis Module
- AI Recommendation Engine
- Health Analytics Dashboard
- MongoDB Cloud Database

Chapter 7

Methodology

7.1 Step 1: User Registration

Users securely create accounts and log into the platform.

7.2 Step 2: Health Data Collection

Users enter health-related information such as height, weight, age, gender, and activity level.

7.3 Step 3: BMI Calculation

$$BMI = \frac{Weight(kg)}{Height(m)^2} \tag{7.1}$$

7.4 Step 4: Metabolism Analysis

For males:

$$BMR = 10W + 6.25H - 5A + 5 \tag{7.2}$$

For females:

$$BMR = 10W + 6.25H - 5A - 161 \tag{7.3}$$

Chapter 8

Implementation

The proposed system is implemented using MERN Stack technologies.

8.1 Frontend Technologies

- HTML
- CSS
- JavaScript
- React.js
- Bootstrap

8.2 Backend Technologies

- Node.js
- Express.js

8.3 Database

- MongoDB

The frontend provides responsive dashboards while backend APIs process healthcare calculations and recommendation generation.

Chapter 9

Experimental Results

The proposed system was tested using multiple healthcare datasets.

9.1 Observations

- Accurate BMI calculation
- Efficient metabolism analysis
- Fast healthcare data processing
- Personalized recommendations generated successfully
- Improved user interaction

Chapter 10

Performance Evaluation

10.1 Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{10.1}$$

10.2 Precision

$$Precision = \frac{TP}{TP + FP} \tag{10.2}$$

10.3 Recall

$$Recall = \frac{TP}{TP + FN} \tag{10.3}$$

10.4 F1-Score

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{10.4}$$

Table 10.1: Performance Analysis

Model	Accuracy	Precision	Recall	F1
HealthPulse AI	96%	95%	94%	94.5%

Chapter 11

Advantages

- Automated healthcare analysis
- AI-based recommendations
- Real-time health monitoring
- Responsive web application
- Secure cloud-based healthcare management
- Interactive healthcare dashboards

Chapter 12

Applications

- Hospitals
- Healthcare Organizations
- Fitness Centers
- Wellness Applications
- Educational Institutions
- Telemedicine Platforms
- Nutrition Consultation Systems

Chapter 13

Future Enhancements

Future enhancements include:

- Wearable Device Integration
- AI Chatbot Support
- Disease Prediction Systems
- IoT-Based Healthcare Monitoring
- Blockchain-Based Healthcare Security
- Mobile Application Development

Chapter 14

Conclusion

The proposed HealthPulse AI system successfully demonstrates the integration of Artificial Intelligence and Full Stack Web Technologies for developing an intelligent healthcare platform capable of automated BMI detection, metabolism analysis, calorie estimation, and personalized healthcare recommendation generation.

The system improves preventive healthcare awareness through intelligent healthcare analytics, secure cloud-based healthcare management, and responsive user interfaces.

Therefore, the proposed HealthPulse AI platform provides an efficient, scalable, secure, and intelligent solution for modern healthcare monitoring and preventive healthcare management.

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